LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various functions of CRO, measurement of voltage, current, phase and frequency
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various functions of CRO, measurement of voltage, current, phase and frequency.
5. Specific Objectives	20.3.1.1 Trainee will be able to explain the construction and operation of DMM & CRO.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of CRO functions, measurement of AC voltage, current, phase, and frequency using probes and DMM; Safety: High voltage caution, proper grounding, and probe compensation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a Cathode Ray Oscilloscope?
(b) How does a Digital Multimeter differ from an Analog Multimeter?
(c) Why is phase angle important in AC circuit analysis?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• CRO Construction and Block Diagram • Function of CRT and Electron Gun • Operation of Horizontal and Vertical Deflection • Role of Triggering and Time Base • Display of Waveforms on Screen	L & Ds	PPT, IB	15 min
(b)	• Measurement of AC Voltage using CRO • Measurement of Current via Shunt Resistance • Determination of Phase Difference • Calculation of Frequency from Time Base • Use of Probes and Grounding Techniques	L & Ds	PPT, PDF	20 min
(c)	• Comparison of DMM and CRO Readings • Error Sources in Voltage and Frequency Measurement • Importance of Probe Compensation • Safety Precautions for High Voltage • Recording Observations in Logbook	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the time base control affects the frequency measurement on a CRO.
(b) Describe the procedure for measuring current using a CRO and a shunt resistor.
(c) What are the key differences between measuring RMS voltage with a DMM versus a CRO?

12. Home Assignment and Follow Up	Prepare a brief report comparing the advantages and limitations of using a Digital Multimeter versus a Cathode Ray Oscilloscope for analyzing complex AC waveforms, referencing concepts from the next lesson on Digital Storage Oscilloscopes.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to Special purpose oscilloscope (Digital storage oscilloscope and sampling CRO)	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min